Stephen Mrenna

Senior Scientist, Fermi National Accelerator Laboratory

Fermilab, Batavia, IL 60510, USA mrenna@fnal.gov smrenna.github.io
0000-0001-8731-160X INSPIRE-HEP Google Scholar pythia.org

Particle physics theorist at Fermilab and one of the primary authors of the PYTHIA event generator, a tool used extensively by theorists and experimentalists in particle, astroparticle, and nuclear physics to interpret particle collision data; member of the CMS Collaboration at the Large Hadron Collider. Current research develops machine-learning methods to improve event-generator models and fit them directly to data (normalizing-flow models of hadronization, simulation-based inference, and autodifferentiation-driven fitting); uses quantum computing to explore simple models of confining theories and learn more about hadronization; and builds agentic AI tools for high-energy-physics workflows and theory tasks.

Career Highlights

- Primary co-author of PYTHIA, the standard simulation tool underpinning LHC data analyses and collider phenomenology worldwide ($> 36,000$ citations to the physics manuals alone; PYTHIA 6.4 is among the most-cited high-energy physics papers of all time).
- 14 years of leadership in CMS event generators, serving as Level-3 Manager of the Generators group through LHC Runs 1 and 2, guaranteeing theoretical fidelity and physics correctness across central simulation for the Higgs boson discovery and CMS physics program.
- Founding organizer of the Monte-Carlo-for-Beyond-the-Standard-Model (MC4BSM) and Tevatron-for-LHC (TeV4LHC) workshop series.
- Co-PI on four current federal awards (U.S. DOE and NSF) spanning agentic AI, ML hadronization, advanced cyberinfrastructure, and quantum simulation.
- Pioneering research program in differentiable event generation, quantum simulation of confining theories, and agentic AI frameworks for particle theory.

ACADEMIC AND RESEARCH POSITIONS

2012–present	**Senior Scientist**, Technology Directorate / Scientific Computing Division / Computing for Science, Fermi National Accelerator Laboratory, Batavia, IL
2022–present	**Adjunct Research Professor**, Department of Physics and Astronomy, University of Cincinnati (joint appointment)
2021–present	**Member / Affiliated Researcher**, Chicago Quantum Exchange (CQE)
2006–2012	**Scientist I**, Computing Division, Fermilab
2001–2006	**Associate Scientist**, Computing Division, Fermilab
2001–2006	**Adjunct Associate Professor**, Department of Physics, University of Michigan (joint appointment)
1998–2001	**Postgraduate Researcher** (1998–2000) and **Assistant Research Scientist** (2001), Institute for High Energy Physics, University of California, Davis
1995–1998	**Postdoctoral Appointee**, Theoretical High Energy Physics Division, Argonne National Laboratory
1993–1995	**Research Fellow**, High Energy Physics, California Institute of Technology
1987–1992	**Research Assistant**, Experimental HEP (1987–88) and Particle Theory (1988–92), The Johns Hopkins University

EDUCATION

1993 **Ph.D., Physics** (theoretical particle physics), The Johns Hopkins University, Baltimore, MD
Thesis: "Signals of new physics in QCD from bottom quark production near the Z boson resonance"
Advisors: G. Domokos and S. Kovesi-Domokos

1988 **M.A., Physics**, The Johns Hopkins University

1986 **B.S. with Honors, Physics**, University of Notre Dame, Notre Dame, IN

HONORS AND FELLOWSHIPS

2025 Breakthrough Prize in Fundamental Physics (shared, as a member of the CMS Collaboration; recognized for generator management and Higgs boson discovery simulation)

2017 Fermilab Exceptional Performance Recognition Award

2013 High Energy and Particle Physics Prize of the European Physical Society (shared, as a member of the CMS Collaboration), for the discovery of a Higgs boson (Level-3 manager of CMS Generators)

1991 Dean's Teaching Fellowship, The Johns Hopkins University

1988–1989 Obert Research Fellowship, The Johns Hopkins University

RESEARCH FUNDING AND PROJECTS

Grants (co-PI)

2026–2027 "AI Agents for HEP Simulations and Analysis Operations," U.S. Department of Energy (RFA DE-FOA-0003612, 14-C, Genesis Mission Phase I); with K. Matchev and S. Gleyzer

2025–2028 "Training hadronization ML models on open data," U.S. Department of Energy Office of High Energy Physics (Award DE-SC0026301, DE-FOA-0003177); with P. Ilten and J. Zupan

2024–2027 "Elements: Automated parametric transformations for rapid particle physics simulations," NSF Office of Advanced Cyberinfrastructure (NSF 22-632); with P. Ilten and J. Zupan

2024–2026 "Collaborative Research: CyberTraining: Pilot: Monte Carlo General Education Network (MC-GEN)," NSF Office of Advanced Cyberinfrastructure (NSF 23-520); with P. Ilten and J. Zupan

2000 Linear Collider Detector Simulation and Physics Studies Grant

Grants (participant)

- SciDAC-4: HEP Data Analytics on HPC (2017–2022)
- SciDAC-4: HPC framework for event generation at colliders (2017–2022)
- Quantum Science Center, Thrust 2: Strong Interactions from Quarks to Nuclei (2020–present)
- QuantISED: HEP ML/Optimization Goes Quantum (2019–2024)

COLLABORATION ROLES AND LEADERSHIP

2000– **Member**, PYTHIA collaboration

2007– **Member**, CMS collaboration

2007–2021 **Physics Generators manager**, CMS Collaboration (Level-3 manager of the Generators group, 2015–2021); led physics validation and generator operations for central Monte Carlo production across LHC Runs 1 and 2, guaranteeing the theoretical correctness and physical fidelity of billions of simulated collision events underpinning the Higgs boson discovery and CMS physics analyses

2014–2017 **Co-convener**, CMS Monte Carlo and Interpretations group

	Internal referee for CMS papers, mainly related to event generators
2015	**Co-organizer**, ATLAS/CMS Dark Matter Forum
2001–2009	**Theory representative**, Tevatron New Phenomena Working Group (joint CDF/DØ/theory, Tevatron Run II); first combined CDF and DØ results
2007–2009	**Editor**, TeV4LHC workshop reports
2004–2006	**Leader**, QCD working group, Tevatron-for-LHC (TeV4LHC) workshop
2001–	**Member**, Coordinated Theoretical-Experimental Project on QCD (CTEQ)
1999	**Co-convener**, QCD Tools group, Run II QCD and Weak Boson Physics Workshop
1998	**Co-convener**, Event Generator group, Run II SUSY/Higgs Workshop

SOFTWARE AND STANDARDS

2000–present	**PYTHIA** (pythia.org), general-purpose event generator: co-author of versions 6.1–6.4 and 8.1–8.3 [Comput. Phys. Commun. **135**, 238 (2001); JHEP **05**, 026 (2006); Comput. Phys. Commun. **178**, 852 (2008) and **191**, 159 (2015); SciPost Phys. Codeb. 8 (2022)]. Primary simulation framework utilized across all LHC experiments (CMS, ATLAS, ALICE, LHCb) and worldwide particle, nuclear, and astroparticle collision experiments ($> 36,000$ citations to the physics manuals). Co-developed automated parton-shower variations [Phys. Rev. D **94**, 074005 (2016)] and hadronization reweighting with automated fragmentation variations [SciPost Phys. **16**, 134 (2024)]
2026	**Vistas**, visualization interface for particle-collision simulations (co-author) [arXiv:2606.19524]
2025	**HEPTAPOD**, orchestration of high-energy-physics workflows by AI agents (co-author) [arXiv:2512.15867]
2023–2024	**Simulation-based inference and variance-reduction codes** (co-author) [SciPost Phys. Codeb. 14 (2023); SciPost Phys. Codeb. 28 (2024)]
2021	**Apprentice**, event-generator tuning (co-author) [EPJ Web Conf. **251**, 03060 (2021)]
2006	**Les Houches Event File** standard format (co-author) [Comput. Phys. Commun. **176**, 300 (2007)]
2004	**Patriot**, archive of Monte Carlo event samples and tools for the CDF and DØ Run II physics programs (developer; presented at CHEP 2004)
2001	**Generic user-process interface** for event generators (Les Houches Accord), co-author [hep-ph/0109068]
1996	**SPythia**, supersymmetric extension of PYTHIA 5.7 (author) [Comput. Phys. Commun. **101**, 232 (1997)]

CONFERENCE AND WORKSHOP ORGANIZATION

2026	Co-organizer (with K. Matchev), Agentic AI workshop, LPC, Fermilab, Sep. 2026, and the LPC initiative “Agentic AI for BSM Event Generation and Analysis” (first organizational meeting, Aug. 2026)
2026	Organizing committee, 2026 Meeting of the APS Division of Particles and Fields (DPF 2026), Fermilab, Jul. 2026
2026	Co-organizer (with C. Bauer, A. Ciavarella, J. Halimeh, and Y. Meurice), “From String Dynamics to Event Generation with Quantum Simulation,” InQubator for Quantum Simulation (IQuS), University of Washington, Seattle, Apr. 2026
2020	Co-organizer (with N. Mohamed, K. Pedro, E. Usai, and others), CMS Open Data Workshop for Theorists at the LPC (remote), Sep.–Oct. 2020

2019 Co-organizer, Analysis Description Languages for the LHC, Fermilab, May 2019

2018 Organizing committee, MC4BSM 2018, IPPP Durham, UK, Apr. 2018

2017 Co-organizer, "(Re)interpreting LHC new physics search results: tools and methods," LPC workshop, Fermilab, Oct. 2017

2017 Organizing committee, 11th MC4BSM workshop, SLAC, May 2017

2015 Co-organizer, joint MC4BSM and LPC "data challenge," Fermilab, May 2015

2015 Organizing committee, 9th MC4BSM workshop (LPC–Theory workshop), Fermilab, May 2015

2012 Organizing committee, 6th MC4BSM workshop, Cornell University, Mar. 2012

2010 Organizing committee, 5th MC4BSM workshop, Niels Bohr Institute, Copenhagen, Apr. 2010

2009 Organizing committee, HEFTI Workshop on Missing Energy Signals at the LHC, UC Davis, Apr. 2009

2009 Organizing committee, MC4BSM-4, UC Davis, Apr. 2009

2008 Co-organizer, CTEQ–LHC Workshop

2008 Co-organizer, CDF Method-2 Workshop

2008 Organizing committee, MC4BSM-3, CERN, Mar. 2008

2007 Organizing committee, 2nd MC4BSM workshop ("Physics at LHC: From Experiment to Theory"), Princeton University, Mar. 2007

2006 Founding organizer, 1st MC4BSM (Monte Carlo Tools for Beyond the Standard Model Physics) workshop, Fermilab, Mar. 2006

2005 Co-organizer (with T. Han and J. Hewett), "Collider Physics: From the Tevatron to the LHC to the Linear Collider," Aspen Center for Physics summer program

2004–2006 Founding organizer (with M. Carena), Tevatron-for-LHC (TeV4LHC) workshop series, Fermilab, BNL, and CERN

2004 Co-convener, Electroweak Symmetry Breaking: theory and phenomenology (parallel session), DPF 2004, Riverside, CA

2002 Co-organizer (with J. Huston), Monte Carlo/Matrix Element Working Group, Fermilab

2000 Co-convener, QCD I session, DPF 2000, Columbus, OH

1997 Co-organizer, Argonne SUSY mini-workshop

Snowmass community studies

2022 Snowmass 2021, Energy Frontier topical groups EF05 (Precision QCD), EF06 (Hadronic Structure and Forward QCD), and EF07 (Heavy Ions):
contributor to the topical-group report [arXiv:2209.14872]

2022 Snowmass 2021: co-editor, community white paper "Event Generators for High-Energy Physics Experiments" [SciPost Phys. **16**, 130 (2024)]

2022 Snowmass 2021: contributor, "Theory, phenomenology, and experimental avenues for dark showers: a Snowmass 2021 report" [Eur. Phys. J. C **82**, 1132 (2022)]

2020 Snowmass 2021, Computational Frontier topical group CompF2 (Theoretical Calculations and Simulation): invited panelist, Event Generators panel, Snowmass Computational Frontier Workshop, Aug. 2020

2001 Snowmass 2001, Precision Higgs Working Group (P1WG2): convener and co-author of the working-group summary report [eConf C010630, P1WG2 (2001)]

2001 Snowmass 2001, American Linear Collider Working Group: contributor, *Linear Collider Physics*

Resource Book for Snowmass 2001

2001 Snowmass 2001: contributor, "The Snowmass Points and Slopes: Benchmarks for SUSY Searches" [Eur. Phys. J. C **25**, 113 (2002)]

1996 Snowmass 1996, Supersymmetry Theory group: subgroup leader

MENTORING AND SUPERVISION

URA Visiting scholars

2026 Rikab Gambhir, University of Cincinnati

2026 Zane Ozzello, University of Iowa (QuPYTH hybrid quantum-classical hadronization)

2026 Blake Senseman, University of Iowa (Rydberg atom quantum simulation of string breaking)

2025 Michael Wilkinson, University of Cincinnati (now Research Scientist, University of Oklahoma)

2022 Manuel Szewc, University of Cincinnati (now Faculty, ICAS, UNSAM, Buenos Aires)

2021–2022 Kenneth Heitritter, University of Iowa (now VP of R&D, qBraid)

Thesis committees

2026 Blake Senseman, PhD Comprehensive Examinee, University of Iowa

2025 Thomas Sievert, PhD Candidacy Examinee, Caltech

2009 Nils Lavesson, External faculty PhD opponent, Lund University, Sweden (now in quantum technology / industry)

2006 Mitch Soderberg, Doctoral committee, University of Michigan (now Professor and Department Chair, Syracuse University)

Other

2026 Ken Wu, Google Summer of Code, University of Waterloo

2020–2025 Prasanth Shyamsundar, mentee, Fermilab Research Associate (now Senior AI Researcher, Fermilab)

2020–2023 Naomi Cooke, MCnet student, University of Birmingham (now Research Fellow in Italy; previously University of Glasgow)

2018–2019 Darleen Perez-Lavin, NSF MSGI, University of Kentucky (now Research Scientist, Naval Information Warfare Center)

TUTORIALS AND LECTURES

CMS Data Analysis Schools

2026 Generators short exercise

2025 Generators short exercise

2022 Generators short exercise

2021 Generators short exercise

2020 Generators short exercise

2019 Generators short exercise

2015 Generators short exercise

2012 Generators short exercise

2011 Generators short exercise

2011 SUSY OSETs exercise

CTEQ Summer Schools

2011	"Monte Carlo and matching," Madison, WI
2009	"Monte Carlo and event generators," Madison, WI
2007	"Event Generators," Madison, WI
2005	"Basics of event generators," Puebla, Mexico
2004	"Test of event generators at the TeV Scale," Madison, WI
2004	"Basics of event generators," Puebla, Mexico
2002	"Introduction to event generators," Puebla, Mexico
2001	"Event generators," St. Andrews, Scotland

Miscellaneous

2026	MCGEN (`https://gitlab.com/mcgen-ct/tutorials`) tutorials at 2026 CFNS Summer School on the Physics of the Electron-Ion Collider (presented by Phil Ilten)
2025	MCGEN (`https://gitlab.com/mcgen-ct/tutorials`) tutorials at CTEQ 2025 Summer School (presented by Phil Ilten)
2024	"Hands-on 2 of PYTHIA 8" (with L. Lönnblad), CMS Generators group tutorials, CERN
2017	PYTHIA 8 Bootcamp (three-part lecture course), Dartmouth–TRIUMF–UW HEP/Cosmology Tools Bootcamp
2011	"The SUSY you need to know for 2011," two-part LPC Topic of the Week lectures, Fermilab
2010	Tutorial on OSETs and "simplified models" techniques, LPCC workshop, CERN; led the follow-up question session with M. Lisanti
2009	"Understanding Cross Sections at the LHC," four lectures, CERN Academic Training Programme
2009	"Physics Monte Carlos," US CMS JTerm III, Fermilab LPC
2004	Lectures on Pythia (with T. Sjöstrand and P. Skands), full-day course for CDF, Fermilab
1991	Instructor, *Neural Networks in Physics* (own design, accredited undergraduate course), The Johns Hopkins University (Dean's Teaching Fellowship)
1986–1991	Graduate teaching assistant, The Johns Hopkins University: recitations and laboratories for introductory physics; problem sessions and substitute lectures in classical and quantum mechanics; shared lecturing for a graduate course in experimental particle physics

OUTREACH AND COMMUNITY ENGAGEMENT

2009–present	**Scouting America** (formerly Boy Scouts of America) Scoutmaster (2020–present); Assistant Scoutmaster and Advancement Chair (2009–2019) Recipient of 12 Eagle Scout Mentor Pins High-adventure advisor: Philmont Scout Ranch (3 times), Boundary Waters (2 times), Swamp Base, Isle Royale Merit badge counselor: Hiking, Backpacking, Fishing
2014–2020	**FIRST LEGO League (FLL) and VEX robotics teams** Coach and mentor Competition referee
Nov 2024	Moderator, World Wide Data Day
Jul 2023	Invited lecture, "Truth and Uncertainty in Particle Physics," Thomistic Philosophy & Natural Science Symposium, Dominican House of Studies, Washington, DC (featured on The Thomistic Institute Podcast)

Jan 2014	Moderator, LHC Masterclass
1993–1998	Science education and community outreach: Partners in Education curriculum design for middle-school physics (table-top electrostatic generator and rail gun, Huntington Middle School, CA); science-fair judge and Argonne National Laboratory educational open house

PROFESSIONAL SERVICE

Committees and panels

2026–present	**J. J. and Noriko Sakurai Dissertation Award Selection Committee**, APS Division of Particles and Fields (DPF); took part in award deliberations and organized a committee meeting
2024–present	**Coordinating Panel on Software and Computing (CPSC)**, APS DPF; member of the inaugural panel (nominated and approved in the 2024–25 cycle)
2023–present	**Scientific Advisory Committee (SAC)**, Fermilab; scientific guidance on Fermilab programmatic activities
2023–present	**Authorship Review Committee (ARC)**, CMS Collaboration; oversight and review of CMS authorship
2010–2012	**Selection committee**, URA Outstanding Doctoral Thesis prize
2007–2009	**Selection committee**, NSF LHC Theory Initiative Fellowship

Refereeing and reviewing

- **Journal referee**: Physical Review Letters, Physical Review D, Physics Letters, Journal of High Energy Physics, Computer Physics Communications, and other APS and HEP journals
- **Grant reviewer**: U.S. National Science Foundation, U.S. Department of Energy, European Research Council

PUBLICATIONS

Theory and phenomenology (papers not signed as CMS Collaboration)

50,190	56	139
citations	h-index	records

All INSPIRE-HEP records, including CMS Collaboration papers

267,566	215	1,706
citations	h-index	records

PYTHIA manuals citations: $>$**36,000**. Listed below: **150** selected publications (**75** refereed).

Source: INSPIRE-HEP, 15 October 2026.

Recent preprints, submitted for publication (10)

[10] G. T. Fleming *et al.* (incl. **S. Mrenna**; 83 authors) [Craig Group], "Blueprint for an AI-Enabled Particle Theory Community," (2026), arXiv:2610.28505. *Submitted for publication.*

[9] B. Assi *et al.* (incl. **S. Mrenna**; 9 authors), "Pythia-Contrib: a modular contribution framework for the Pythia 8 event generator," (2026), arXiv:2610.19111. *Submitted for publication.*

[8] T. Menzo, A. Roman, G. T. Fleming, S. Gleyzer, K. Matchev and **S. Mrenna**, "HEPTABench: A framework for portable and controlled agentic evaluations in high energy physics," (2026), arXiv:2610.18454. *Submitted for publication.*

[7]T. Menzo, G. T. Fleming, K. T. Matchev, **S. Mrenna** and A. Roman, "Reining in an Agentic Harness for High Energy Physics," (2026), arXiv:2609.00107. *Submitted for publication.*

[6]B. Senseman, Z. Ozzello, Y. Meurice and **S. Mrenna**, "Confinement, String Breaking, and Hadronization in the Compact Abelian Higgs Model," (2026), arXiv:2607.27550. *Submitted for publication.*

[5]B. Assi *et al.* (incl. **S. Mrenna**; 10 authors), "Vistas: A Visualization Interface for Particle Collision Simulations," (2026), arXiv:2606.19524. *Submitted for publication.*

[4]T. Menzo, A. Roman, G. T. Fleming, S. Gleyzer, K. T. Matchev and **S. Mrenna**, "Agentic Diagrammatica: Towards Autonomous Symbolic Computation in High Energy Physics," (2026), arXiv:2603.26990. *Submitted for publication.*

[3]T. K. Aarrestad *et al.* (incl. **S. Mrenna**; 459 authors), "Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision," (2026), arXiv:2602.17582. *Submitted for publication.*

[2]T. Menzo *et al.* (incl. **S. Mrenna**; 8 authors), "HEPTAPOD: Orchestrating High Energy Physics Workflows Towards Autonomous Agency," (2025), arXiv:2512.15867. *Submitted to SciPost Physics Codebases.*

[1]N. Heller *et al.* (incl. **S. Mrenna**; 8 authors), "Rejection Sampling with Autodifferentiation - Case study: Fitting a Hadronization Model," (2024), arXiv:2411.02194. *Submitted for publication.*

Most-cited publications (also included in the lists below)

T. Sjostrand, **S. Mrenna** and P. Z. Skands, "PYTHIA 6.4 Physics and Manual," JHEP **05**, 026 (2006), arXiv:hep-ph/0603175. 13,747 citations

T. Sjostrand, **S. Mrenna** and P. Z. Skands, "A Brief Introduction to PYTHIA 8.1," Comput. Phys. Commun. **178**, 852 (2008), arXiv:0710.3820. 8,416 citations

T. Sjöstrand *et al.* (incl. **S. Mrenna**; 10 authors), "An introduction to PYTHIA 8.2," Comput. Phys. Commun. **191**, 159 (2015), arXiv:1410.3012. 8,338 citations

T. Sjostrand *et al.* (incl. **S. Mrenna**; 7 authors), "High-energy physics event generation with PYTHIA 6.1," Comput. Phys. Commun. **135**, 238 (2001), arXiv:hep-ph/0010017. 3,376 citations

C. Bierlich *et al.* (incl. **S. Mrenna**; 14 authors), "A comprehensive guide to the physics and usage of PYTHIA 8.3," SciPost Phys. Codeb. **2022**, 8 (2022), arXiv:2203.11601. 1,851 citations

B. Allanach *et al.* (incl. **S. Mrenna**; 35 authors), "The Snowmass Points and Slopes: Benchmarks for SUSY Searches," Eur. Phys. J. C **25**, 113 (2002), arXiv:hep-ph/0202233. 967 citations

D. Alves *et al.* (incl. **S. Mrenna**; 95 authors) [LHC New Physics Working Group], "Simplified Models for LHC New Physics Searches," J. Phys. G **39**, 105005 (2012), arXiv:1105.2838. 958 citations

T. Sjostrand, L. Lonnblad and **S. Mrenna**, "PYTHIA 6.2: Physics and manual," (2001), arXiv:hep-ph/0108264. 876 citations

D. Abercrombie *et al.* (incl. **S. Mrenna**; 140 authors), "Dark Matter benchmark models for early LHC Run-2 Searches: Report of the ATLAS/CMS Dark Matter Forum," Phys. Dark Univ. **27**, 100371 (2020), arXiv:1507.00966. 803 citations

J. Alwall *et al.* (incl. **S. Mrenna**; 35 authors), "A Standard format for Les Houches event files," Comput. Phys. Commun. **176**, 300 (2007), arXiv:hep-ph/0609017. 699 citations

Refereed journal articles (75)

[75]A. Butter *et al.* (incl. **S. Mrenna**; 13 authors), "Iterative HOMER with uncertainties," SciPost Phys. **20**, 042 (2026), arXiv:2509.03592.

[74]B. Senseman, Z. Ozzello, K. Heitritter, Y. Meurice and **S. Mrenna**, "Prolegomena to a hybrid classical-Rydberg simulator for hadronization," Phys. Rev. D **114**, 036017 (2026), arXiv:2212.02476.

[73]A. Al Kadhim, H. B. Prosper and **S. Mrenna**, "Bayesian optimization of pythia8 tunes," Phys. Rev. D **112**, 056012 (2025), arXiv:2505.11675.

[72]B. Assi *et al.* (incl. **S. Mrenna**; 9 authors) [MLhad], "Post-hoc reweighting of hadron production in the Lund string model," SciPost Phys. **19**, 104 (2025), arXiv:2505.00142.

[71]Y. Kyselov, **S. Mrenna** and M. Ovchynnikov, "New physics particles mixing with mesons: Production in the fragmentation chain," Phys. Rev. D **112**, 055033 (2025), arXiv:2504.06828.

[70]C. Agapopoulou *et al.* (incl. **S. Mrenna**; 25 authors) [HEP Software Foundation], "The Critical Importance of Software for HEP," Eur. Phys. J. C **85**, 1142 (2025), arXiv:2504.01050.

[69]B. Assi *et al.* (incl. **S. Mrenna**; 9 authors), "Characterizing the hadronization of parton showers using the HOMER method," SciPost Phys. **19**, 125 (2025), arXiv:2503.05667.

[68]C. Bierlich *et al.* (incl. **S. Mrenna**; 8 authors), "Describing hadronization via histories and observables for Monte-Carlo event reweighting," SciPost Phys. **18**, 054 (2025), arXiv:2410.06342.

[67]N. Cooke, P. Ilten, L. Lönnblad and **S. Mrenna**, "Non-relativistic quantum chromodynamics in parton showers," Eur. Phys. J. C **84**, 432 (2024), arXiv:2312.05203.

[66]C. Bierlich *et al.* (incl. **S. Mrenna**; 8 authors), "Towards a data-driven model of hadronization using normalizing flows," SciPost Phys. **17**, 045 (2024), arXiv:2311.09296.

[65]P. Shyamsundar, J. L. Scott, **S. Mrenna**, K. T. Matchev and K. Kong, "Variance Reduction via Simultaneous Importance Sampling and Control Variates Techniques Using Vegas," SciPost Phys. Codeb. **2024**, 28 (2024), arXiv:2309.12369.

[64]C. Bierlich *et al.* (incl. **S. Mrenna**; 8 authors), "Reweighting Monte Carlo predictions and automated fragmentation variations in Pythia 8," SciPost Phys. **16**, 134 (2024), arXiv:2308.13459.

[63]J. Campbell *et al.* (incl. **S. Mrenna**; 212 authors), "Event generators for high-energy physics experiments," SciPost Phys. **16**, 130 (2024), arXiv:2203.11110.

[62]S. Frixione, S. Amoroso and **S. Mrenna**, "Matrix element corrections in the Pythia8 parton shower in the context of matched simulations at next-to-leading order," Eur. Phys. J. C **83**, 970 (2023), arXiv:2308.06389.

[61]A. C. Li, A. Macridin, **S. Mrenna** and P. Spentzouris, "Simulating scalar field theories on quantum computers with limited resources," Phys. Rev. A **107**, 032603 (2023), arXiv:2210.07985.

[60]C. Bierlich *et al.* (incl. **S. Mrenna**; 14 authors), "A comprehensive guide to the physics and usage of PYTHIA 8.3," SciPost Phys. Codeb. **2022**, 8 (2022), arXiv:2203.11601.

[59]G. Albouy *et al.* (incl. **S. Mrenna**; 54 authors), "Theory, phenomenology, and experimental avenues for dark showers: a Snowmass 2021 report," Eur. Phys. J. C **82**, 1132 (2022), arXiv:2203.09503.

[58]A. Macridin, A. C. Li, **S. Mrenna** and P. Spentzouris, "Bosonic field digitization for quantum computers," Phys. Rev. A **105**, 052405 (2022), arXiv:2108.10793.

[57]S. Höche, **S. Mrenna**, S. Payne, C. T. Preuss and P. Skands, "A Study of QCD Radiation in VBF Higgs Production with Vincia and Pythia," SciPost Phys. **12**, 010 (2022), arXiv:2106.10987.

[56]W. Wang *et al.* (incl. **S. Mrenna**; 8 authors), "BROOD: Bilevel and Robust Optimization and Outlier Detection for Efficient Tuning of High-Energy Physics Event Generators," SciPost Phys. Core **5**, 001 (2022), arXiv:2103.05751.

[55]A. Valassi *et al.* (incl. **S. Mrenna**; 39 authors) [HSF Physics Event Generator WG], "Challenges in Monte Carlo Event Generator Software for High-Luminosity LHC," Comput. Softw. Big Sci. **5**, 12 (2021), arXiv:2004.13687.

[54]A. P. Austin, M. Krishnamoorthy, S. Leyffer, **S. Mrenna**, J. Müller and H. Schulz, "Practical algorithms for multivariate rational approximation," Comput. Phys. Commun. **261**, 107663 (2021), arXiv:1912.02272.

[53]J. Alimena *et al.* (incl. **S. Mrenna**; 201 authors), "Searching for long-lived particles beyond the Standard Model at the Large Hadron Collider," J. Phys. G **47**, 090501 (2020), arXiv:1903.04497.

[52]A. Boveia *et al.* (incl. **S. Mrenna**; 33 authors), "Recommendations on presenting LHC searches for missing transverse energy signals using simplified s-channel models of dark matter," Phys. Dark Univ. **27**, 100365 (2020),

arXiv:1603.04156.

[51]D. Abercrombie *et al.* (incl. **S. Mrenna**; 140 authors), "Dark Matter benchmark models for early LHC Run-2 Searches: Report of the ATLAS/CMS Dark Matter Forum," Phys. Dark Univ. **27**, 100371 (2020), arXiv:1507.00966.

[50]J. S. Gainer *et al.* (incl. **S. Mrenna**; 15 authors), "Adding pseudo-observables to the four-lepton experimentalist's toolbox," JHEP **10**, 073 (2018), arXiv:1808.00965.

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INVITED TALKS AND PRESENTATIONS

Invited Conference Presentations, Plenaries, and Seminars

Sep 2026	*AI for Theory* — plenary presentation, CLARIPHY Collaboration Meeting, University of Wisconsin–Madison
Sep 2026	*Community Vision* — Agentic AI for BSM Event Generation and Analysis Workshop, LPC, Fermilab
Apr 2026	*Hadronization Models in Event Generation* — IQuS workshop "From String Dynamics to Event Generation with Quantum Simulation," University of Washington, Seattle
Jun 2025	*Studying Hadronization with ML and the Road to "differentiable" Pythia* — CIPANP 2025, Madison, WI
Oct 2023	*Event Generators and Pythia8* — High Energy Theory Seminar, Cincinnati
Oct 2023	*Uncertainty, LHC Event Generators, and Pythia8* — INT Workshop "Theoretical Physics Uncertainties to Empower Neutrino Experiments" (INT-23-86W), Institute for Nuclear Theory, Seattle
Feb 2020	*Why and How is MC tuned to data?* — LPC Physics Forum, Fermilab
Nov 2019	*Physics generators and the community roadmap for HEP in the next decade* — Latin American Workshop on Software and Computing Challenges in HEP (LAWSCHEP 2019)
Jun 2019	*Analysis Description Languages for High Energy Physics* — minisymposium presentation (with S. Sekmen), Platform for Advanced Scientific Computing (PASC19) Conference, ETH Zürich, Switzerland
May 2019	*Roundtable: mass measurements, MC, supportive measurements* — Top Quark Physics at the Precision Frontier, Fermilab
May 2019	*Summary / How to move forward* — Analysis Description Languages for the LHC, Fermilab
Jan 2018	*Theoretical introduction: mass* — Top Quark Physics at the Precision Frontier, Fermilab
Oct 2017	*The Pythia8 Event Generator for Particle Collisions: Status and Updates* — HEP Seminar, Tufts

Sep 2017	*The Pythia8 Event Generator for Particle Collisions: Status and Updates* — Particle Physics Seminar, Notre Dame
May 2017	*Experimental requirements – CMS* — MC4BSM 2017, SLAC
Apr 2017	*Status of the PYTHIA8 Event Generator* — Pittsburgh Particle physics, Astrophysics and Cosmology Center (PITT PACC), University of Pittsburgh
Sep 2016	*Pythia LO Generator: Software Operation and Organization* — Beyond Leading Order Calculations on HPCs, Fermilab
May 2015	*Overview of CMS–ATLAS joint dark matter working group* — and *MC4BSM/LPC Data Challenge*, joint MC4BSM and LPC "data challenge," Fermilab
May 2014	*M3: Pythia* — MC4BSM-2014, Daejeon, South Korea
Mar 2012	*Fast Detector Simulation* — MC4BSM-2012 (6th MC4BSM workshop), Cornell University
Oct 2011	*Simplified models in CMS* — LPCC workshop "Implications of LHC results for TeV-scale physics," WG2, CERN
Apr 2011	*Panel Discussion on Hard SUSY Models* — SUSY Recast Workshop, UC Davis
Jan 2011	*The Calculable and the Observable* — Boston Jet Physics Workshop, Harvard University
Nov 2010	*Implementing simplified models in PYTHIA, and parameter space sampling* — Characterization of New Physics at the LHC II, CERN
Nov 2009	*Tools for Calculations and Simulations* — Muon Collider Workshop, Fermilab
Sep 2009	*What we need for the LHC: the CMS point of view* — Northwest Terascale Research Projects: $W + b$ quark physics at the LHC, Univ. of Oregon
Mar 2009	*Monte Carlo Tools for Early Physics* — LHC at BNL Workshop, Brookhaven
Feb 2009	*Parton Distributions for Parton Showers* — Northwest Terascale Research Projects: Parton Showers and Event Structure at the LHC, Univ. of Oregon
Mar 2008	*MC for BSM in CMS* — MC4BSM-3, CERN
Feb 2008	*MC author's perspective* — PDF4LHC meeting, CERN
Jan 2008	*How Well Do We Understand the Standard Model?* — Aspen Winter Conference "Revealing the Nature of Electroweak Symmetry Breaking"
Mar 2007	*How well do we understand high-p_T data?* — Physics at LHC: From Experiment to Theory (MC4BSM-2), Princeton University
Mar 2007	*The Status of Monte Carlo Tuning* — and *Using the Vista of high-p_T data to validate event generators*, 3rd HERA and the LHC workshop, DESY Hamburg
Jul 2006	*Bard: Interpreting Frontier Physics* — Harvard
Jun 2006	*Preparing for Discovery* — SUSY06
Apr 2006	*Tevatron Measurements for the LHC* — Rencontres de Moriond (Electroweak)
Mar 2006	*Bard: Interpreting Frontier Physics* — 1st MC4BSM workshop, Fermilab
Jan 2006	*LHC Phenomenology: A Bard's Eye View* — Aspen Winter Workshop
Aug 2005	*TeV4LHC* — Aspen Summer Workshop on Collider Physics
May 2005	*Monte-Carlo Tools for TeV Colliders* — Les Houches Workshop: Physics at TeV Colliders
Apr 2005	*Understanding the Backgrounds to Top* — Michigan Center for Theoretical Physics Top Quark Symposium
Aug 2004	*New Phenomena: Models, Parameters, and Formats* — The Tevatron Connection, Fermilab
Apr 2004	*Pythia for the LHC Era* — ATLAS-Japan Workshop on Physics Simulation for LHC, Tsukuba

Feb 2004	*Merging Matrix Elements and Parton Showers at Hadron Colliders* — KITP Program: Collider Physics
Jul 2003	*Implementation issues in Pythia* — Monte Carlo Tools for the LHC: Matrix Element tools, CERN
May 2002	*Event Generators and Higher-Order Corrections* — LoopFest, Brookhaven (plenary)
May 2001	*SUSY Higgs at the Tevatron* — Workshop on the Future of Higgs Physics, Fermilab (plenary)
Oct 2000	*Pinning Down the MSSM Higgs Sector for Large or Small Phases* — 5th International Linear Collider Workshop, Fermilab
Aug 2000	*The Reach of the LHC for SUSY models with a Degenerate Wino Pair* — DPF 2000, Columbus
May 2000	*Mapping the SUSY New World* — Argonne SUSY Institute (divisional seminar)
Apr 2000	*SUSY in Run I, Run II, and Run II++* — SUSY Tools Workshop 2000, Colmar (plenary)
Mar 2000	*The Challenges of Finding SUSY With a Wino or Higgsino LSP/NLSP Pair* — Berkeley 2000 Meeting, LBL (plenary)
Dec 1999	*The Complementarity of the Tevatron and LHC for the Standard Model-like Higgs bosons* — PASCOS 99, Granlibakken (plenary)
Jun 1999	*The Complementarity of the LHC and the Tevatron for probing the Standard Model-like Higgs* — and *Searching for SUSY at the Tevatron when $\widetilde{C}_1$ and $\widetilde{N}_1$ are degenerate*, SUSY99, Fermilab
Apr 1999	*Technicolor in Pythia* — Strong Dynamics Workshop, Fermilab
Apr 1999	*Corrections to Parton Showering from Resummation* — Tools for SUSY Workshop, Lyon
Mar 1999	*Corrections to Parton Showering from Resummation Calculations* — Fermilab Theory Seminar
Mar 1999	*Parton Showering at Next-to-Leading Order* — QCD and Weak Boson Physics Workshop, Fermilab
Oct 1998	*The Strawman Model of Technicolor in Pythia* — Strong Dynamics Workshop, Fermilab
Oct 1998	*Top Quark Decays beyond the Standard Model: What to Expect* — Top Thinkshop, Fermilab

Selected Collaboration and Working Group Presentations

Oct 2026	*AI Policy in* PYTHIA *Software Development* — HSF Software Technology Forum "Responsible AI-assisted coding and common training aspects," CERN
Sep 2026	*Tuning: What, How, and Who* — theory talk, CMS Physics Days: Physics modelling and MC generators, CERN
Sep 2025	*Developments in fragmentation and hadronization models* — CMS Standard Model Physics (SMP) General Meeting, CERN
Jan 2024	*Parton shower modeling and systematics for top quarks* — Top Physics Analysis Group Workshop, CERN
May 2021	*Pythia8 report for LHCC review* — HSF Physics Event Generator WG Meeting #13
Sep 2020	*The Dead Cone in PYTHIA 8* — ALICE substructure analyses subgroup
Feb 2020	*PYTHIA 8.3 Update* — CMS/Scientific Computing meeting, Fermilab
Jun 2019	*Thoughts on constraining modelling systematic uncertainties* — CMS Top Physics group mini-workshop on systematic uncertainties, CERN
Dec 2018	*Outcomes and Future Activities* — HSF Generator Software Workshop organisation meeting
Jun 2018	*PYTHIA tunes* — LHC Electroweak WG: Jets and EW bosons, CERN
May 2017	*Thoughts on Reducing FSR Uncertainties for Top Decays* — CMS Top Quark Physics group meeting, CERN
Jan 2017	*Parton shower uncertainty calculations and DIRE plugin* — CMS SMP-J annual workshop, CERN

Jan 2016	*PYTHIA 8 Status Report* — ATLAS–CMS Monte Carlo Generators Workshop, CERN
Nov 2015	*Strategies for Discovery: Feedback from theorists* — CMS Exotica Workshop, Venice
Nov 2014	*Dark Matter models and EXO plans for run2* — CMS SUSY Event at the LPC, Fermilab
Oct 2014	*Simplified models and generation* — CMS Run 2 Monojet Mini-Workshop, CERN
Feb 2014	*Thoughts on being a theorist in CMS* — CMS Collaboration Board, CERN
Nov 2013	*CMS status of MC tuning* — QCD Tools for LHC Physics: From 8 to 14 TeV, Fermilab
Jun 2012	*Overview of unique sensitivity* — CMS SUSY Workshop, DESY
May 2011	*Theory Perspectives* — Topology Workshop: Hadronic Final States, CMS Physics Week, CERN
Oct 2005	*Phenomenological Overview of Modeling Uncertainties* — CDF/DØ Top Mass Workshop
Apr 2005	*W + jets at the Tevatron: the story so far* — TeV4LHC meeting, CERN
Apr 2005	*Understanding W/Z+jets from Theory* — Joint CTEQ/CDF/DØ Workshop on W/Z Physics, Fermilab
Dec 2004	*FSR/ISR effects on $h \to b\bar{b}$ production* — Higgs Working Group, TeV4LHC workshop, Fermilab
Oct 2004	*Patriot: Physics Archives and Tools Required to Investigate Our Theories* — CDF Joint Physics meeting, Fermilab; and CHEP 2004, Interlaken (poster)
Jan 2004	*High p_T Physics Simulation* — Joint CDF/DØ/CMS Jet/MET Workshop, Fermilab